

Computational Reproduction Record

This repository record preserves the complete R46 and R47 computational narratives and implementation protocols, and supplies all R48 per-instance metrics. Mathematical statements and proofs remain in the current article and electronic companion. Antecedent measurements are not relabeled as new runs.

CR.1 Computational Evidence for Joint Design

The primary protocol fixes 24 independent-seed synthetic cases with two, four, or eight input branches; nine or seventeen catalog points; two or three installed commands; and zero or nonnegative heterogeneous charges. Eight additional cases vary the raw history count and response-type count using a 33-point catalog. Levels lie in $[0, 1]$, the reward has $r = 2$ and $q = 1$, and shortfall curvatures are drawn from $\{1, 2, 4, 8\}$. Participation caps, weights, and eligibility ceilings vary independently within the declared rational generator. The protocol was frozen before these runs, not externally preregistered. Every generated case is retained. Larger comparator extensions were added after the primary run and cover the entire declared comparison set, not selected successes.

The proposed method uses eight root price-bisection steps, retains price zero, and uses the same budget and charges in every incumbent and bound. Primary box runs have a 301-node and eight-second soft limit; repeated-type scaling runs have a 101-node limit. Time is checked after completing a node, so some recorded times exceed eight seconds. The only initial incumbent is a fixed-target conditional book at the capped ideal targets, followed by exact allocation for that book. It is not a global certificate. The baseline price-prefix search reimplements the inherited R43 exact completion bound, uses the same price budget, and stops at the same absolute tolerance. A weaker full-catalog prefix relaxation is reported separately, not substituted for this stronger baseline.

Table 2 reports 28 of 32 box runs attaining absolute gap 0.001, versus 32 of 32 for the priced-prefix baseline. The box method has lower measured optimization time in 11 cases, but it is not uniformly faster or more accurate. Median times are 0.093 seconds for boxes and 0.070 seconds for priced prefixes. Four box runs retain unresolved gaps between 0.00217 and 0.01248. These are valid intervals, not achieved-tolerance claims. The median Jensen-minus-incumbent interval is 0.01124; the median final box interval is 0.00018. Every box interval passes the independent structural checker.

The eight smallest holdout cases also have independent all-book joint references. Their incumbent losses are zero in this sample, but the priced bounds need not be exact. This observation is not a guarantee for the initial target choice or any heuristic portfolio. Repeated-type cases with 128 and 256 histories reduce exactly to four or eight response types; their certificates are lifted and checked on the original history space. The sixteen-type cases have no such large reduction. These rows demonstrate full joint intervals after a stated exact aggregation, not merely the runtime of conditional target compression.

Table 3 tests the complete-grid reference at successively smaller tolerances and with two through five branches. Only the three two-branch grid runs complete within the 1,000-profile limit. The adaptive method completes eleven of twelve corresponding requests; its four-branch, 0.001 run remains interrupted. Exact-reference losses, normalized objective scales, and all counts are provided rather than inferring accuracy from

sampled objective agreement. This comparison tests the new coverage mechanism, not a claim that generic spatial partitioning is novel.

The independent mixed-integer formulation retains installation binaries, branch support indicators, probabilities, pre-draw service, the root equality, and every realization implication. Its tangent and secant models are solved on all 24 primary cases with 64 cost segments and separate four-second limits. All 48 solves return optimal numerical status. Table EC.4 gives each case’s runtimes, search nodes, primal–dual gaps, envelope error and bracket width. Negative signed widths at floating roundoff scale are retained in the raw records and are not called exact certificates. The results show that generic MIP and inherited prefix search remain competitive on these catalogs. The methodological claim for the box oracle is exact lower-bound-aware decomposition and independently checkable adaptive joint certificates, not universal computational superiority.

Regression includes 600 exact all-book price comparisons on heterogeneous target boxes, 24 further joint interval comparisons, 2,793 independent finite-lattice checks of constructive net coverage, genuine below-anchor and above-cap clipping, the one-branch one-call case, and eleven rejected certificate mutations. The three-target descending-side example in Section 5 is an exact counterexample to reusing a rising-only oracle on a target box. The coverage checker is structurally independent of the search routine. These are verification tests, not statistical evidence about an operational population.

The complete historical numerical study remains in Section CR.5. Its thirty-five-of-thirty-six portfolio result and safeguarded sample exactness are not used as the present global-algorithm headline. Its inserted-ideal-target continuous example is an oracle candidate-pool sanity check, and its large conditional table is explicitly labeled “targets fixed.” No customer records, estimated installation charges, field calibration, or deployment outcomes are asserted.

CR.2 Independent Formulation and Numerical Error Bounds

The independent optimizer does not import the canonical response, either path recurrence, or the fixed-book allocation code. Let $z_i \in \{0, 1\}$ enable level a_i , let $w_{ji} \in \{0, 1\}$ indicate a permitted branch support, let $p_{ji} \geq 0$ be its probability, and let $y_j \in [0, b_j]$ be the intermediate tier chosen before the draw. The original quadratic catalog problem is represented by

$$\max \quad \sum_j \pi_j \left[\sum_i p_{ji} f(a_i) - \frac{\gamma_j}{2} y_j^2 \right] - \sum_i \rho_i z_i, \quad (\text{CR.1})$$

$$\text{s.t.} \quad \sum_i z_i \leq s, \quad 0 \leq p_{ji} \leq w_{ji} \leq z_i, \quad \sum_i p_{ji} = 1, \quad (\text{CR.2})$$

$$y_j + \sum_i a_i p_{ji} \leq b_j, \quad \sum_j \pi_j \left(y_j + \sum_i a_i p_{ji} \right) = B, \quad (\text{CR.3})$$

$$y_j + a_i \leq \tau_j + M_{ji}(1 - w_{ji}), \quad M_{ji} = \max\{0, b_j + a_i - \tau_j\}. \quad (\text{CR.4})$$

For $a_i > \tau_j$, set $w_{ji} = p_{ji} = 0$. The bound $y_j \leq b_j$ makes the displayed big- M sufficient. No adjacent-support constraint is imposed. A feasible contract gives a feasible point by activating its positive support and installed levels. Conversely every model point defines one pre-draw tier and a lottery obeying both the target cap and every realized ceiling. Thus the model represents the original feasible set, not its canonical reduction. The allowance of activated levels with zero use does not change the optimum under nonnegative charges: redundant activations can be removed.

For the numerical comparison, replace $\gamma_j y^2/2$ by an epigraph of piecewise-affine functions. With nodes $x_r = r b_j/K$, $0 \leq r \leq K$, the tangent envelope is

$$h_j^{\tan}(y) = \max_{0 \leq r \leq K} \{\gamma_j x_r y - \gamma_j x_r^2/2\}.$$

The secant envelope is the maximum of the K consecutive chord lines. On $[0, b_j]$,

$$0 \leq h_j - h_j^{\tan} \leq \frac{\gamma_j b_j^2}{8K^2}, \quad 0 \leq h_j^{\sec} - h_j \leq \frac{\gamma_j b_j^2}{8K^2}.$$

The tangent formula follows by completing the square and choosing the nearest node; the secant formula follows from the quadratic chord identity. Consequently, writing $E_K = \sum_j \pi_j \gamma_j b_j^2/(8K^2)$, the exact envelope optima satisfy

$$V_s \leq V_s^{\tan} \leq V_s + E_K, \quad V_s - E_K \leq V_s^{\sec} \leq V_s.$$

A secant-model feasible policy is feasible for the original constraints, and evaluating its actual quadratic cost can only improve its objective. In the implementation, however, the solver and its dual bounds are floating point. We therefore retain status, reported mixed-integer gap, residuals, raw probabilities, intermediate tiers, and both envelope results; we do not promote a numerical dual bound to an exact rational certificate. Comparing against exact catalog values is a separate validation step. The chosen numerical tolerances and segment count are stored with every result.

CR.3 Independent Rational Policy Checks

For a fixed target and proposed book, the independent exhaustive routine enumerates singleton and all two-point supports directly. For a singleton $a \leq t$, service is $y = t - a$. For a pair $u < v$, the terminal mean is $t - y$ and the high probability is $(t - y - u)/(v - u)$. Feasible service lies in

$$\max\{0, t - v\} \leq y \leq \min\{t - u, \tau - v\}.$$

Its payoff is the pair's affine reward at $t - y$ minus the nondecreasing intermediate cost. Because the chord slope is positive, the smallest feasible y maximizes this payoff. Enumerating singleton and pair supports therefore gives a direct branch optimum without invoking the canonical implementation. Concavity and the standard two-moment extreme-point argument justify that no larger support is required. Catalog-subset enumeration then gives a separate exact check of conditional book selection.

The policy checker verifies that each activated level belongs to the catalog, the book is strictly ordered and within its stated budget, charges are paid once, each probability is nonnegative, and each lottery sums to one. It recomputes each target as $y_j + \sum_i a_i p_{ji}$, the exact root equality, every cap, and every positive-probability realization $y_j + a_i \leq \tau_j$. It separately recomputes the quadratic reward and intermediate cost. Semantic mutation cases change targets, probabilities, support, service, charges, promise, or the declared budget; the test record identifies each rejection. This checker validates a policy and conditional enumeration. It does not infer global target optimality from a feasible policy or from the same price recurrence.

CR.4 Target Nets, Approximation Errors, and Support Stability

The net implementation fixes a possible lowest installed level, enumerates $k - 1$ weighted target coordinates, fills the remaining coordinate subject to its cap, and repairs any positive residual by bounded increases. It chooses a branch of largest probability as the distinguished coordinate, solely to avoid unnecessary enumeration; the theorem does not depend on this choice. The code stores the weighted mesh step, evaluated profile count, anchor count, requested accuracy, and completion status. A profile cap can interrupt execution, but then the global ε guarantee is explicitly absent. All returned policies still satisfy the resource and contractual constraints.

In floating-point implementations, aggregate equality error alone does not bound all target perturbations. Let t^U be the exact mixed target, and let $\tilde{t}$ be its rounded version after clipping to the chosen book's feasible target intervals. Compute or upper-bound

$$r_{\text{mix}} = \sum_j \pi_j |\tilde{t}_j - t_j^U|, \quad r_{\text{eq}} = \left| B - \sum_j \pi_j \tilde{t}_j \right|.$$

After exact residual repair, the weighted distance from t^U is at most $r_{\text{mix}} + r_{\text{eq}}$. Hence the safe additional value allowance in the compressed-support theorem is $L(r_{\text{mix}} + r_{\text{eq}})$, not merely a bound on the equality residual. Clipping is included in r_{mix} . Canonical reconstruction is performed after repair, and any desired rational probability representation is generated from the repaired targets and exact installed levels. Independently rounding probabilities and then checking only their sum would not establish root or risk feasibility.

The theory does not require continuity of the selected book. Tied optima or perturbed parameters can change several selected levels while the corresponding certified value interval changes little. In particular, uncertain eligibility thresholds must be replaced by declared safe bounds before optimization; a lottery feasible for an estimated threshold is not automatically safe under downward perturbation. The error theorem concerns certified oracle and target arithmetic at stated model data. It is not an unproved robustness result for unknown customer parameters.

CR.5 Current Reproduction and Preserved Evidence

The exact command budget counts selected, enabled levels. It is not the number of levels receiving positive probability after optimization; zero-charge exact-cardinality frontiers may contain unused selections. The

original archival column “Used” is a selected-level count. The current code reports the book and the positive-probability support separately through the full lottery record.

The primary protocol, all original inputs, and completed and interrupted outcomes are retained in `PROTOCOL.json` and `results/`. Run `code/tests.py`, `code/benchmark.py`, `code/comparators.py`, and `code/price_compare.py` to regenerate scientific evidence; `code/tables.py` generates the displayed values. Benchmarks record actual wall time and platform. A resumed run retains completed records rather than selecting new inputs. The comparator extension covers all primary cases, and the priced-prefix extension covers all 32 cases. They were not part of an external preregistration. The separate `check_certificate.py` verifies a supplied rational target-box certificate without importing the search routine.

The complete R45 main paper and electronic companion are preserved in `predecessor/`, alongside their source entry points. Their historical computational sections, protocols, and complete tables remain available there and in their unchanged source directories, rather than being appended again as new experimental evidence. Every mathematical theorem, proposition, lemma, corollary, and proof from the current R45 readers remains in the current article or this companion. The content map identifies each current, moved, or archived input. All inherited repository paths outside the explicitly replaced root readers and index files are byte-preserved.

In the archived charged portfolio exception, the exact same-budget value, the portfolio value and their difference are recorded together in the new response, so the loss 101/3600 is not presented without scale. The archive’s interior target-adaptive exact example includes the ideal targets in its pool; it is an oracle-pool sanity check. Its 32–256-branch timing table optimizes the book conditional on supplied targets and is not pooled with the present joint intervals.

CR.6 Joint Computation with Distinct Histories

The additional protocol contains twelve primary cases with four, eight, or twelve distinct histories and twelve scaling cases with 32, 128, or 256 distinct histories. Primary catalogs have nine levels and budgets two or three; scaling catalogs have seventeen levels and budget three. Four cap centers and four cost centers receive rational, history-specific perturbations at scales 0.001 and 0.02. We vary nonuniform weights, nonnegative charges, one or four eligibility regimes, and interior or saturated promises. Every cap–cost pair is distinct, so the exact response-type reduction leaves every history separate. This is a declared clustered synthetic family, not a generic random population. The protocol and generator hash were fixed before these runs, not externally preregistered.

Coarsening uses cap and curvature bins twice the perturbation scale, exact eligibility classes, and one minimum-cap guard. It gives four groups for four-history cases and five otherwise. These practical bins are deliberately coarser than the conservative accuracy prescription in Theorem 7.3; their validity comes from the checked original-space interval, not from assuming a small worst-case error. The inner target-box tolerance is 0.0005, and the requested original tolerance is 0.001. Both box methods use eight root-price steps, a 101-node limit and a four-second soft limit. No original reallocation improvement is added after the groupwise lift. The priced-prefix baseline uses the same original tolerance and resource limits. Soft limits

are checked between complete oracle blocks; recorded times exceeding them are retained.

Tables 4 and 5 report 24 of 24 original-space certificates within tolerance for coarsening, 19 for unaggregated boxes, and 21 for priced prefixes. Median optimization times are 0.037, 0.318, and 0.169 seconds, respectively. The largest coarsened gap is 0.000783. This is a global certificate result, not an exact-hit claim about a target portfolio. All 48 coarsened and raw intervals pass independent semantic checking; full original policies and tree witnesses are supplied. The inherited 32-case study, including its four unresolved box runs and stronger prefix results, remains intact in Section CR.1. Its runtimes are not pooled with this execution.

All twelve primary cases have exhaustive all-book joint references. That enumeration is independent of the target search but shares the exact fixed-book allocator; it is not presented as a separately derived allocation algorithm. The largest lifted-incumbent loss to those references is $3.03e-13$. A genuinely uneliminated original-model mixed-integer formulation provides a separate check: its 24 tangent/secant solves retain every support implication and do not use canonical allocation. Table EC.2 reports each solver status, time, node count, gap, envelope error, and signed numerical bracket width. These floating-point bounds are not substituted for rational certificates.

An additional four-case stress protocol was fixed after the primary run to challenge coarse recovery, rather than to extend its success rate. In Table EC.1, the reduced models are solved exactly but initial original-space gaps range from 27/640 to 67/100. None is mislabeled as meeting tolerance. One refinement to original singletons closes every gap to zero. This demonstrates both the need to account for aggregation error and the constructive route back to the original problem; it does not claim that one refinement always suffices.

New regression tests include 72 exhaustive original/reduced optimum pairs, 602 fixed-book lifts, 441 cap-dispersion checks, 216 accuracy-binned partitions, and 56 exact refinement comparisons. Genuine clipping at saturation restores every individual cap. The independent checker rejects 18 semantic mutations, including changed representatives, lost branches, altered charges, false gaps, and missing coverage leaves. Prior tests and derivations are preserved without relabeling them as new runs. The study establishes algorithmic and certificate behavior on these declared inputs; it does not assert field calibration or universal runtime superiority.

References

- Aardal K, van den Berg P, Gijswijt D, Li S (2014) Approximation algorithms for hard capacitated k -facility location problems. Author manuscript, arXiv:1311.4759v4, September 12.
- Aggarwal A, Klawe MM, Moran S, Shor P, Wilber R (1987) Geometric applications of a matrix-searching algorithm. *Algorithmica* 2:195–208. doi:10.1007/BF01840359.
- Chan TM (2021) (Near-)linear-time randomized algorithms for row minima in Monge partial matrices and related problems. *Proceedings of the 2021 ACM–SIAM Symposium on Discrete Algorithms*, 1465–1482. doi:10.1137/1.9781611976465.88.
- Croci M, Fasi M, Higham NJ, Mary T, Mikaitis M (2022) Stochastic rounding: Implementation, error analysis and applications. *Royal Society Open Science* 9:211631. doi:10.1098/rsos.211631.
- Del Pia A, Dey SS, Molinaro M (2017) Mixed-integer quadratic programming is in NP. *Mathematical Programming* 162:225–240. doi:10.1007/s10107-016-1036-0.
- Eppstein D (2005) PADS: Python Algorithms and Data Structures. *SMAWK.py*, offline matrix-search implementation, originally written 2002 and revised 2005. University of California, Irvine. MIT-licensed distribution; source version and excerpt are identified in the code supplement.
- Fattahi A (2025) Utilities’ managed home charging programs for electric vehicles. *Management Science* 71(12):10205–10224. doi:10.1287/mnsc.2023.00850.
- Fu M (2024) A tutorial on quantized feedback control. *IEEE/CAA Journal of Automatica Sinica* 11(1):5–17. doi:10.1109/JAS.2023.123972.
- Grønlund A, Larsen KG, Mathiasen A, Nielsen JS, Schneider S, Song M (2017) Fast exact k -means, k -medians and Bregman divergence clustering in 1D. Preprint, arXiv:1701.07204, version 4.
- Handler GY, Zang I (1980) A dual algorithm for the constrained shortest path problem. *Networks* 10(4):293–309. doi:10.1002/net.3230100403.
- Huangfu Q, Hall JAJ (2018) Parallelizing the dual revised simplex method. *Mathematical Programming Computation* 10(1):119–142. doi:10.1007/s12532-017-0130-5.
- Hübner T, Gupte A, Rebennack S (2025) Spatial branch-and-bound for nonconvex separable piecewise linear optimization. *INFORMS Journal on Computing*, published online June 27. doi:10.1287/ijoc.2024.0755.
- Jourdain B, Pagès G (2021) Optimal dual quantizers of 1D log-concave distributions: Uniqueness and Lloyd like algorithm. *Journal of Approximation Theory* 267:105581. doi:10.1016/j.jat.2021.105581.
- Kaplan H, Mozes S, Nussbaum Y, Sharir M (2017) Submatrix maximum queries in Monge matrices and partial Monge matrices, and their applications. *ACM Transactions on Algorithms* 13(2), Article 26. doi:10.1145/3039873.

- Klawe MM, Kleitman DJ (1990) An almost linear time algorithm for generalized matrix searching. *SIAM Journal on Discrete Mathematics* 3(1):81–97. doi:10.1137/0403009.
- Lawler EL, Wood DE (1966) Branch-and-bound methods: A survey. *Operations Research* 14(4):699–719. doi:10.1287/opre.14.4.699.
- Lemaréchal C (2001) Lagrangian relaxation. Jünger M, Naddef D, eds. *Computational Combinatorial Optimization*, Lecture Notes in Computer Science 2241 (Springer), 112–156. doi:10.1007/3-540-45586-8_4.
- Li Y, Bertsekas D (2025) An error bound for aggregation in approximate dynamic programming. Author manuscript, arXiv:2507.01324v1, July 2.
- Liao W, Wu J (2015) Matrix inequalities for the difference between arithmetic mean and harmonic mean. *arXiv:1501.04823*.
- Nielsen F, Nock R (2014) Optimal interval clustering: Application to Bregman clustering and statistical mixture learning. Author manuscript, arXiv:1403.2485.
- Pagès G, Wilbertz B (2012) Intrinsic stationarity for vector quantization: Foundation of dual quantization. Author manuscript, arXiv:1010.4642v2, March 27. Original preprint 2010.
- Spear SE, Srivastava S (1987) On repeated moral hazard with discounting. *Review of Economic Studies* 54(4):599–617. doi:10.2307/2297484.
- Stouffer K, Pease M, Tang C, Zimmerman T, Pillitteri V, Lightman S, Hahn A, Saravia S, Sherule A, Thompson M (2023) Guide to operational technology (OT) security. NIST Special Publication 800-82 Rev. 3. doi:10.6028/NIST.SP.800-82r3.
- Thomas J, Worrall T (1988) Self-enforcing wage contracts. *Review of Economic Studies* 55(4):541–554. doi:10.2307/2297404.
- Virtanen P, Gommers R, Oliphant TE, et al. (2020) SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods* 17:261–272. doi:10.1038/s41592-019-0686-2.
- Wu X (1991) Optimal quantization by matrix searching. *Journal of Algorithms* 12(4):663–673. doi:10.1016/0196-6774(91)90039-2.
- Yao FF (1980) Efficient dynamic programming using quadrangle inequalities. *Proceedings of the Twelfth Annual ACM Symposium on Theory of Computing*, 429–435. doi:10.1145/800141.804691.
- Zhang H (2012) Solving an infinite horizon adverse selection model through finite policy graphs. *Operations Research* 60(4):850–864. doi:10.1287/opre.1120.1056.
- Zuse Institute Berlin (2026) PySCIPOpt documentation: Non-linear expressions. Official documentation, accessed September 24. <https://pyscipopt.readthedocs.io/en/latest/tutorials/expressions.html>.

CR.7 Complete Paired Outcomes

All cases in the declared protocol are listed. H and M denote harmonic and minimum-curvature aggregation. C and U denote a certified tolerance and an unresolved original-space gap. Times include reduction and lift but exclude independent checking. The objective scale is $r + \Gamma/2$ plus the largest possible selected charge; scaled gaps divide by that stated scale, not by a possibly zero or negative optimum.

Table CR.1: Every original-space interval and execution cost.

Case	k	d	m	H gap	M gap	H sec.	M sec.	H/M	H nodes	Scale
case-00	4	2	1	7.273e-05	0.005223	0.007067	0.01756	C/U	1	3.303
case-01	6	3	3	0.01757	0.01798	0.09332	0.1154	U/U	23	3.511
case-02	8	4	2	0.001003	0.008414	0.008683	0.005686	U/U	1	6.004
case-03	4	4	3	2.196e-05	2.196e-05	0.0151	0.01344	C/C	1	6.044
case-04	6	2	1	0	0.005836	0.006112	0.005672	C/U	1	3.486
case-05	8	3	3	0.02529	0.02522	0.1212	0.093	U/U	31	3.628
case-06	4	4	2	0	0	0.007206	0.005672	C/C	1	6.004
case-07	6	4	3	7.878e-06	7.878e-06	0.0155	0.0134	C/C	1	6.048
case-08	8	2	1	0.0005123	0.009995	0.008584	0.02171	C/U	1	3.593
case-09	4	3	3	0.0008874	0.0008874	0.12	0.1181	C/C	31	3.333
case-10	6	4	2	0.003184	0.009106	0.008	0.005815	U/U	1	6.003
case-11	8	4	3	0.0005434	0.0004811	0.1744	0.1697	C/C	31	6.044
case-12	32	2	1	1.242e-05	0.0001593	0.01876	0.008649	C/C	1	3.884
case-13	32	4	3	0.02647	0.02651	0.2471	0.1906	U/U	31	3.929
case-14	32	4	2	0.002726	0.004005	0.02676	0.009981	U/U	1	6.005
case-15	32	4	3	0.0004471	3.402e-05	0.03163	0.02185	C/C	1	6.05
case-16	128	2	1	1.152e-05	0.0001534	0.08403	0.01338	C/C	1	3.97
case-17	128	4	3	0.02906	0.02903	0.4054	0.1266	U/U	31	4.015
case-18	128	4	2	0.003184	0.004612	0.2144	0.0153	U/U	1	6.005
case-19	128	4	3	4.955e-05	5.621e-05	0.06063	0.02612	C/C	1	6.02
case-20	256	2	1	4.111e-06	0.0001508	0.3307	0.01909	C/C	1	3.985
case-21	256	4	3	0.0316	0.03165	1.095	0.1777	U/U	29	4.025
case-22	256	4	2	0.003552	0.004935	0.7878	0.02204	U/U	1	6.005
case-23	256	4	3	0	7.392e-07	0.141	0.02952	C/C	1	6.025

CR.8 Independent Numerical Mixed-Integer Comparisons

Every primary case has tangent and secant solves of the uneliminated original model. Status 0 means completed; status 1 means a solver limit. Times, nodes and relative primal–dual gaps below are tangent/secant

pairs. The approximation bound is the maximum one-sided quadratic-envelope error. Width is the tangent dual upper bound minus the secant feasible original payoff. These are floating-point diagnostics, never substitutes for the exact semantic certificates.

Table CR.2: All direct MIP comparisons; no timeout or failed solve is omitted.

Case	Status	Seconds	Nodes	MIP gaps	Env. error	Width
case-00	0/0	0.002927/0.00213	1/1	0/0	0.0001154	5.226e-05
case-01	0/0	0.008477/0.007788	1/1	0/0	6.078e-05	0
case-02	0/0	0.02646/0.02367	1/1	0/0	0.00014	6.545e-05
case-03	0/0	0.005721/0.005395	1/1	0/0	0.0001162	0
case-04	0/0	0.002528/0.002488	1/1	0/0	9.626e-05	4.048e-05
case-05	0/0	0.005747/0.005584	1/1	0/0	9.181e-05	0
case-06	0/0	0.008634/0.009637	1/1	0/0	0.0001127	1.759e-05
case-07	0/0	0.02879/0.0256	2/1	0/0	8.268e-05	-3.109e-14
case-08	0/0	0.003193/0.003005	1/1	0/0	0.0001198	9.469e-05
case-09	0/0	0.007799/0.008086	1/1	0/0	7.96e-05	0
case-10	0/0	0.02637/0.02744	1/1	0/0	9.436e-05	1.515e-05
case-11	0/0	0.01151/0.01303	1/1	0/0	9.175e-05	1.11e-16

CR.9 Interpretation and Reproduction

The twelve all-book references independently enumerate menu choices but share the exact fixed-book allocator. The separate MIP retains support indicators, probabilities, pre-draw service and every realized ceiling. The harmonic checker imports neither harmonic reduction nor the search implementation: it rebuilds harmonic means, verifies eligibility and the anchor guard, checks all original lottery constraints, verifies clipping comparison identities and group supporting prices, and delegates the full target-cover and Bellman upper witness to the antecedent independent checker.

The paired study deliberately uses one wide curvature bin within each cap and eligibility group. It is an ablation of the optimistic representative, not an experiment claiming that conservative accuracy bins are always small. The supplied records separately count both accuracy partitions at three tolerances. The final metric pass uses the antecedent node-accounting MIP wrapper; the cases and limits did not change from the initial authoring run. Both local and remote executions must identify their own measured times rather than substitute one environment’s times for another’s. Full rational policies, raw solver messages, scaled-gap denominators, check times, protocol hashes and generators are retained in the revision directory.